PALETTE STUDIO

Python Tkinter Color Palette Application

PROJECT OVERVIEW

Palette Studio is a desktop application for exploring coordinated color palettes. Users can choose from five preset themes, generate a random set of colors, and click any swatch to copy its hexadecimal color code.

AIM

To design and develop an interactive graphical user interface using Python Tkinter that presents color palettes, lets the user switch themes or generate random colors, and copies a selected color code to the clipboard.

KEY FEATURES

- Five curated palettes: Sunset, Ocean, Forest, Candy, and Lavender.
- Random palette generation with five vivid colors.
- Clickable color swatches that copy their hex code.
- Status text confirms the current palette or copied value.

PROCEDURE

1. Import Tkinter for the interface and random for color generation.

2. Define the named color palettes and the application color scheme.

3. Create the fixed-size main window and add a title and short instruction.

4. Build theme buttons that load each preset palette.

5. Display each color as a clickable swatch with its hexadecimal value.

6. Connect each swatch to the clipboard action and update the status message.

7. Generate five random RGB values and show them as a temporary palette.

8. Load the default Sunset palette and start Tkinter's event loop.

HOW TO USE

Run the Python file in an environment with Tkinter available. Choose a theme button to view its colors, select a color swatch to copy the hex value, or use **Surprise Me** to create a random palette.

PROGRAM CODE

The complete source is reproduced below. The listing continues on the following pages.

```
import tkinter as tk
import random

PALETTES = {
    "SUNSET": ["#FF6B6B", "#F7B267", "#F79D65", "#C8553D", "#723D46"],
    "OCEAN": ["#03045E", "#0077B6", "#00B4D8", "#90E0EF", "#CAF0F8"],
    "FOREST": ["#283618", "#606C38", "#A3B18A", "#DDA15E", "#BC6C25"],
    "CANDY": ["#FF99C8", "#FCF6BD", "#D0F4DE", "#A9DEF9", "#E4C1F9"],
    "LAVENDER": ["#6A4C93", "#9D84B7", "#C8B6E2", "#EDE3F5", "#F7F1FA"],
}

BG = "#171923"
CARD_BG = "#242735"
TEXT = "#F7F7FB"
MUTED = "#A6A9B8"

root = tk.Tk()
root.title("Palette Studio")
root.geometry("760x570")
root.configure(bg=BG)
root.resizable(False, False)

current_colors = []

def copy_color(color):
    root.clipboard_clear()
    root.clipboard_append(color)
    status_label.config(text=f"{color} copied!")

def show_palette(theme, colors=None):
    global current_colors
    current_colors = list(colors if colors else PALETTES[theme])

    theme_label.config(text=f"{theme} PALETTE")
    status_label.config(text="Click a color card to copy its hex code.")

    for widget in color_frame.winfo_children():
        widget.destroy()

    for color in current_colors:
        card = tk.Frame(color_frame, bg=CARD_BG, padx=7, pady=7)
        card.pack(side="left", padx=7)

        swatch = tk.Button(
            card,
            text="",
            bg=color,
            activebackground=color,
```

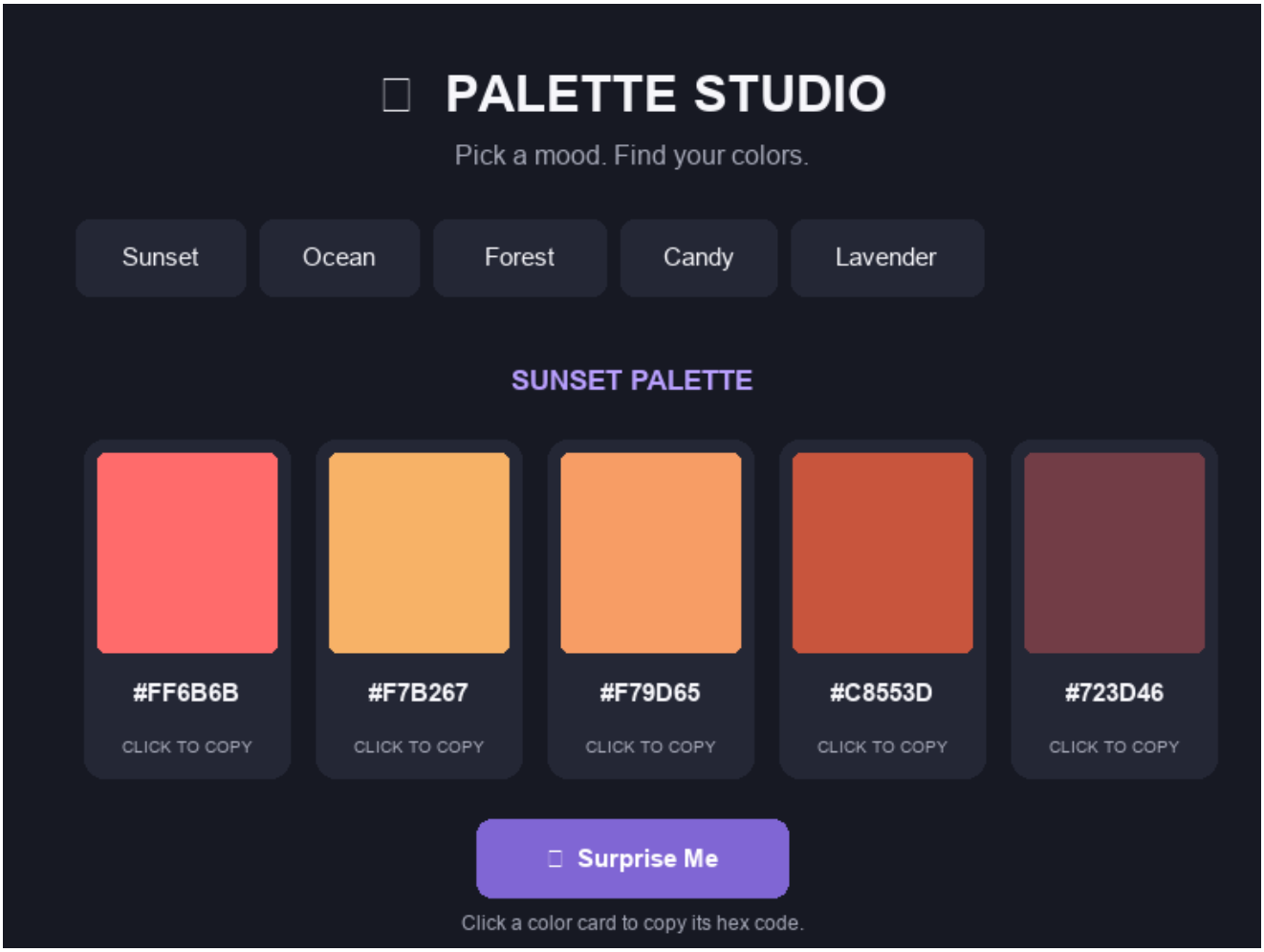
```
            width=12,
            height=7,
            relief="flat",
            cursor="hand2",
            command=lambda c=color: copy_color(c)
        )
        swatch.pack()

        tk.Label(
            card,
            text=color,
            bg=CARD_BG,
            fg=TEXT,
            font=("Arial", 10, "bold")
        ).pack(pady=(9, 2))

        tk.Label(
            card,
            text="CLICK TO COPY",
            bg=CARD_BG,
            fg=MUTED,
            font=("Arial", 7)
        ).pack()

def random_palette():
    colors = [
        "#{:02X}{:02X}{:02X}".format(
            random.randint(40, 240),
            random.randint(40, 240),
            random.randint(40, 240)
        )
        for _ in range(5)
    ]
    show_palette("YOUR RANDOM", colors)

tk.Label(
    root,
    text="❖  PALETTE STUDIO",
    bg=BG,
    fg=TEXT,
    font=("Arial", 25, "bold")
).pack(pady=(25, 5))

tk.Label(
    root,
    text="Pick a mood. Find your colors.",
    bg=BG,
    fg=MUTED,
```

```
    font=("Arial", 11)
).pack(pady=(0, 20))

theme_frame = tk.Frame(root, bg=BG)
theme_frame.pack()

for theme in PALETTES:
    tk.Button(
        theme_frame,
        text=theme.title(),
        bg=CARD_BG,
        fg=TEXT,
        activebackground="#393D50",
        activeforeground=TEXT,
        relief="flat",
        padx=13,
        pady=8,
        cursor="hand2",
        command=lambda t=theme: show_palette(t)
    ).pack(side="left", padx=5)

theme_label = tk.Label(
    root,
    text="",
    bg=BG,
    fg="#B9A0FF",
    font=("Arial", 12, "bold")
)
theme_label.pack(pady=(30, 12))

color_frame = tk.Frame(root, bg=BG)
color_frame.pack()

tk.Button(
    root,
    text="✦  Surprise Me",
    bg="#8066D4",
    fg="white",
    activebackground="#6C53BD",
    activeforeground="white",
    font=("Arial", 11, "bold"),
    relief="flat",
    padx=20,
    pady=10,
    cursor="hand2",
    command=random_palette
).pack(pady=27)
```

```
status_label = tk.Label(
    root,
    text="",
    bg=BG,
    fg=MUTED,
    font=("Arial", 10)
)
status_label.pack()

show_palette("SUNSET")
root.mainloop()
```

OUTPUT

Preview of the application showing the default Sunset palette.



RESULT

Palette Studio was implemented as a Tkinter desktop application. It provides preset and randomized color palettes, clickable swatches, clipboard copying, and on-screen status feedback. The project demonstrates GUI construction, event handling, functions, loops, dictionaries, and random number generation.